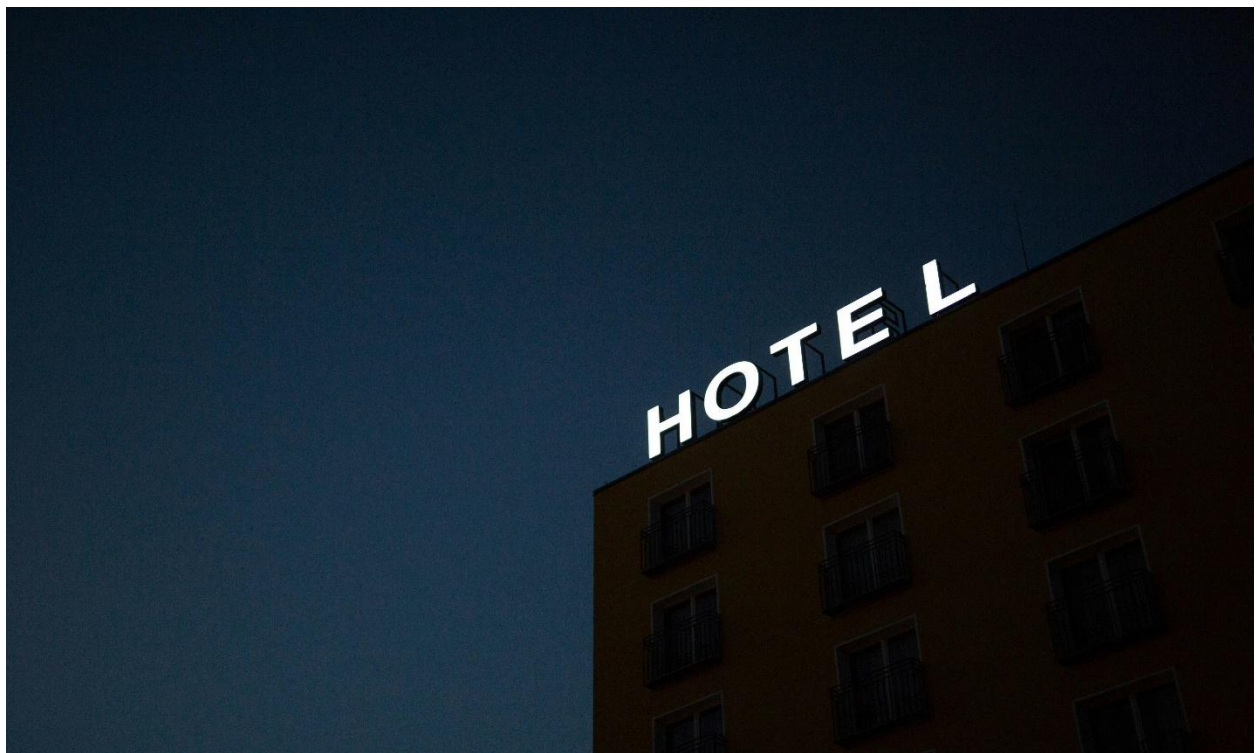


Hotel Booking Analysis Report

Business Problem

In recent years, both **City Hotel** and **Resort Hotel** have experienced high **cancellation rates**. These cancellations have led to several challenges, including reduced revenues and suboptimal utilization of hotel rooms. Lowering the cancellation rate is now a primary goal for both hotels to improve revenue generation and operational efficiency.

This analysis focuses on understanding the drivers behind hotel booking cancellations, as well as examining other factors that may impact business performance and annual revenue generation. The ultimate aim is to provide actionable business recommendations to address these challenges.



Assumptions

1. **Data Stability:** No extraordinary events occurred between 2015 and 2017 that significantly impacted the dataset.

2. **Data Relevance:** The data remains relevant for drawing insights and making future plans for the hotels.
3. **Implementation Neutrality:** There are no unforeseen negative consequences associated with implementing the recommended strategies.
4. **Current Practices:** The hotels have not yet adopted any of the proposed solutions.
5. **Primary Impact:** Booking cancellations are the main factor affecting the hotels' revenue generation effectiveness.
6. **Room Vacancy:** Cancellations directly lead to vacant rooms for the originally booked period.
7. **Booking Timing:** Guests tend to cancel reservations within the same year as their booking.

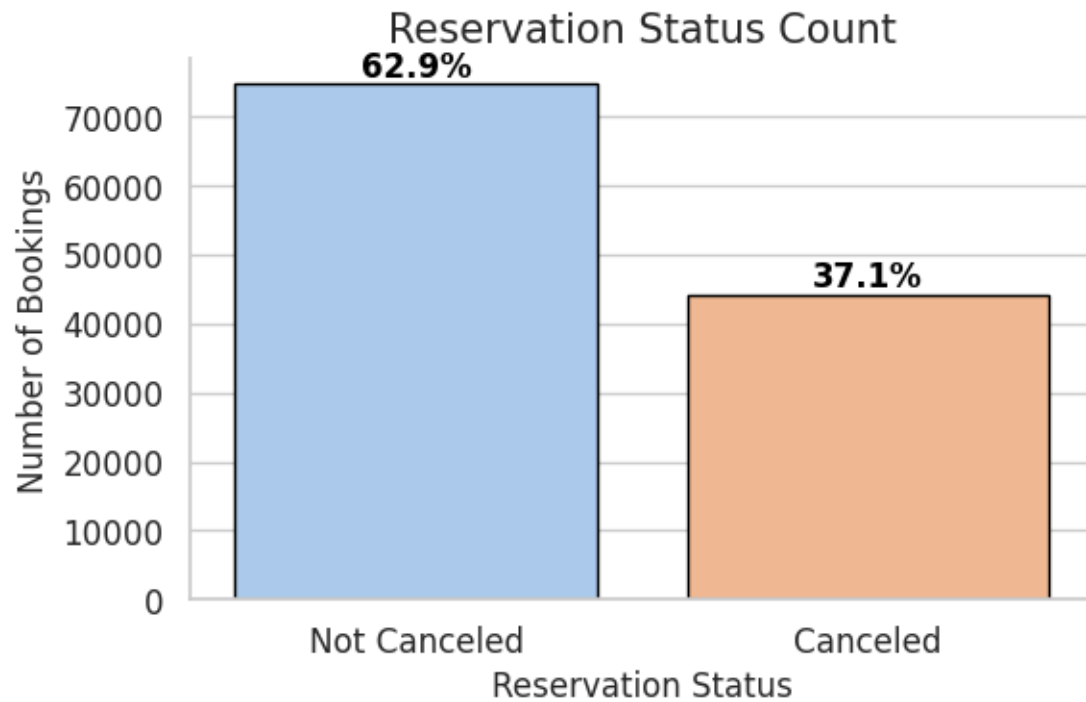
Research Questions

1. What are the key variables influencing hotel reservation cancellations?
2. What actionable strategies can be proposed to reduce cancellation rates?
3. How can insights from the analysis assist hotels in making effective pricing and promotional decisions?

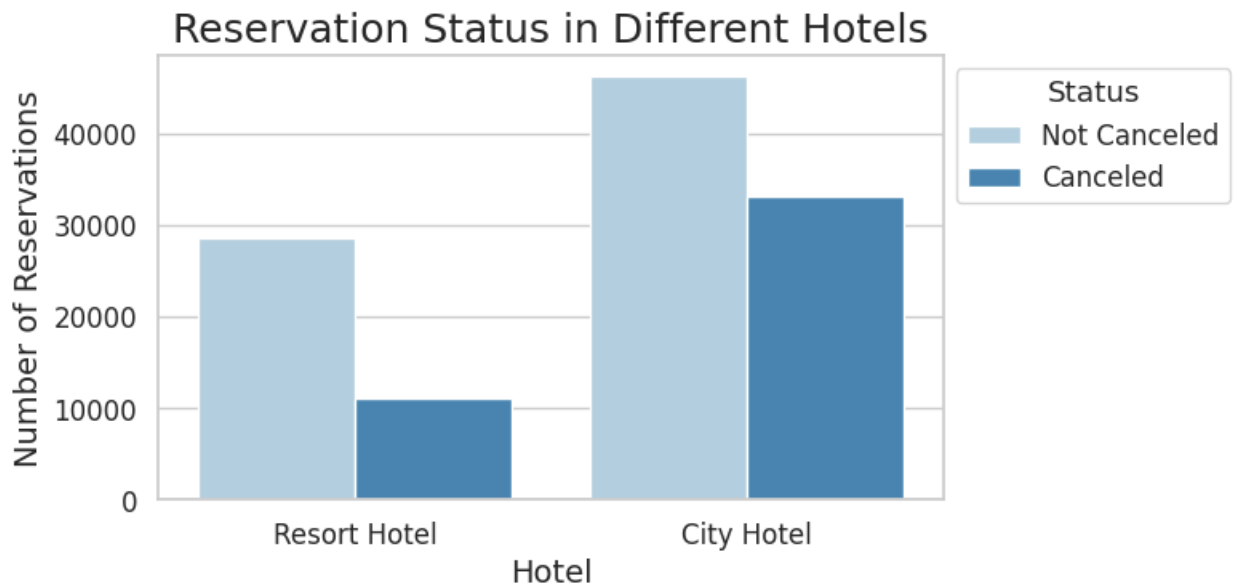
Hypothesis

1. More cancellations occur when prices are higher.
2. When there is a longer waiting list. Customers tend to cancel more frequently.
3. The majority of clients are coming from offline travel agents to make their reservations.

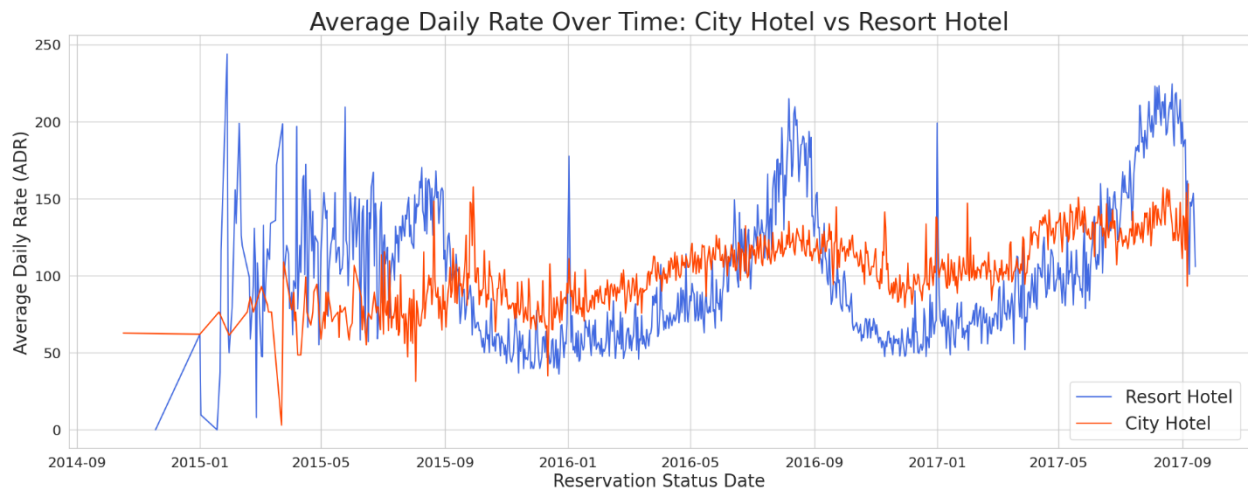
Analysis and Findings



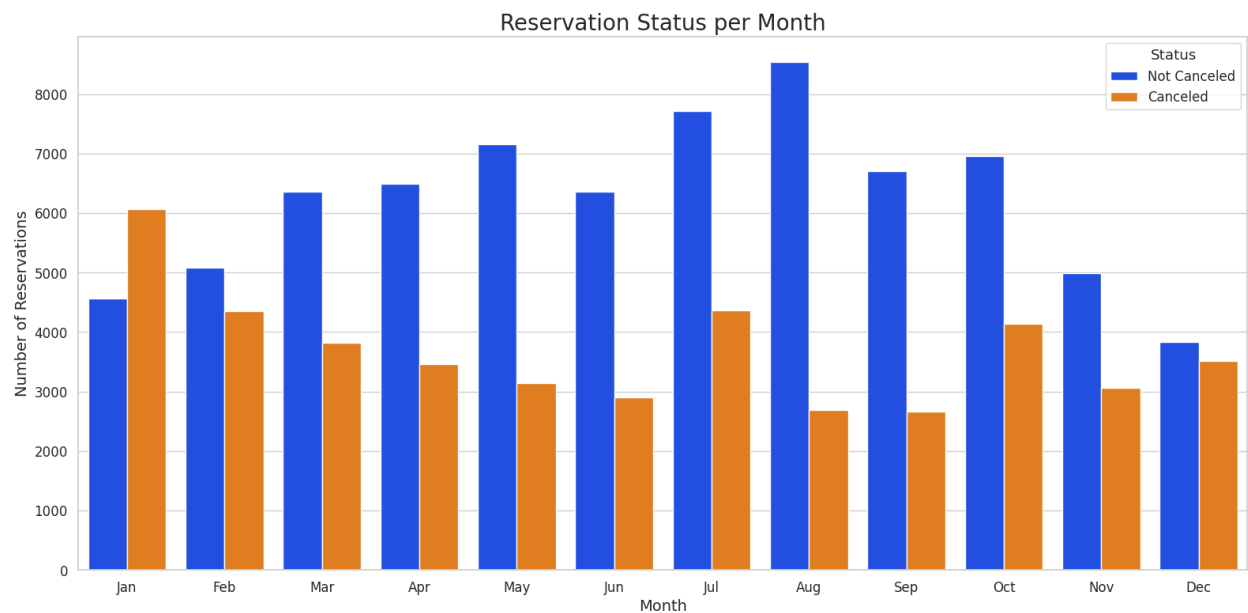
The accompanying bar graph shows the percentage of reservations that are canceled and those that are not. It is obvious that there are still a significant number of reservations that have not been canceled. There are still 37% of clients who canceled their reservation, which has a significant impact on the hotels' earnings.



In comparison to resort hotels, city hotels have more bookings. It's possible that resort hotels are more expensive than those in cities.

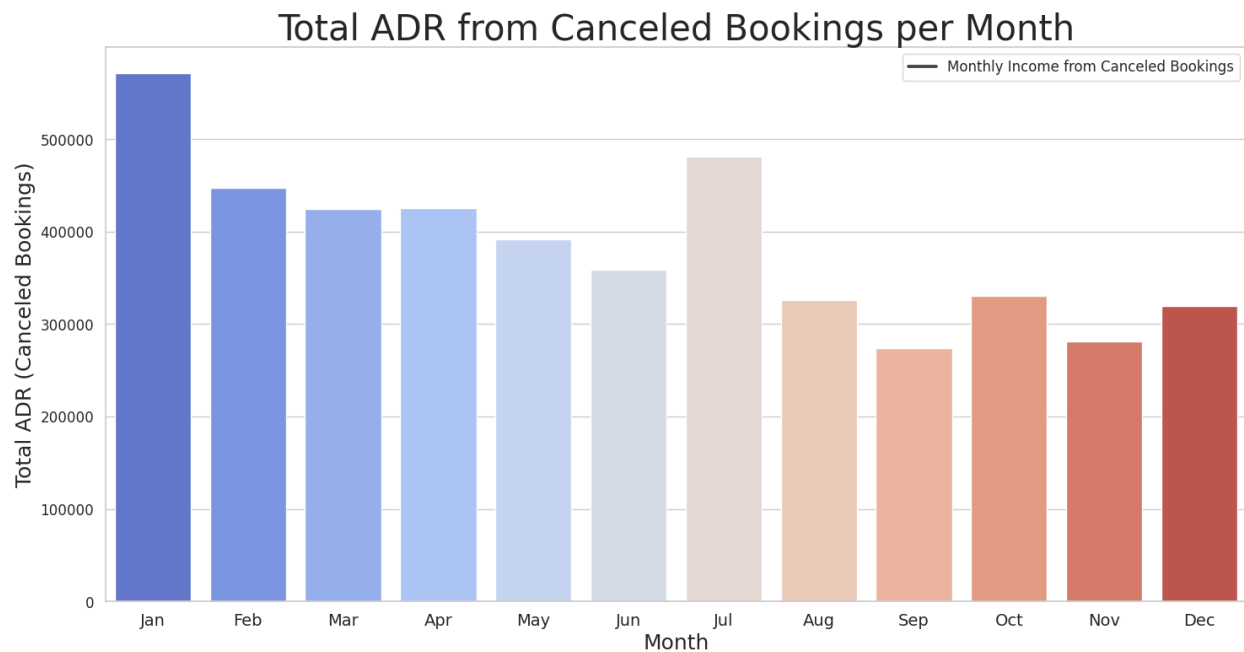


The line graph above shows that, on certain days, the average daily rate for a city hotel is less than that of a resort hotel, and on other days, it is even less. It goes without saying that weekends and holidays may see a rise in resort hotel rates.



We have developed the grouped bar graph to analyze the months with the highest and lowest reservation levels according to reservation status. As can be seen, both

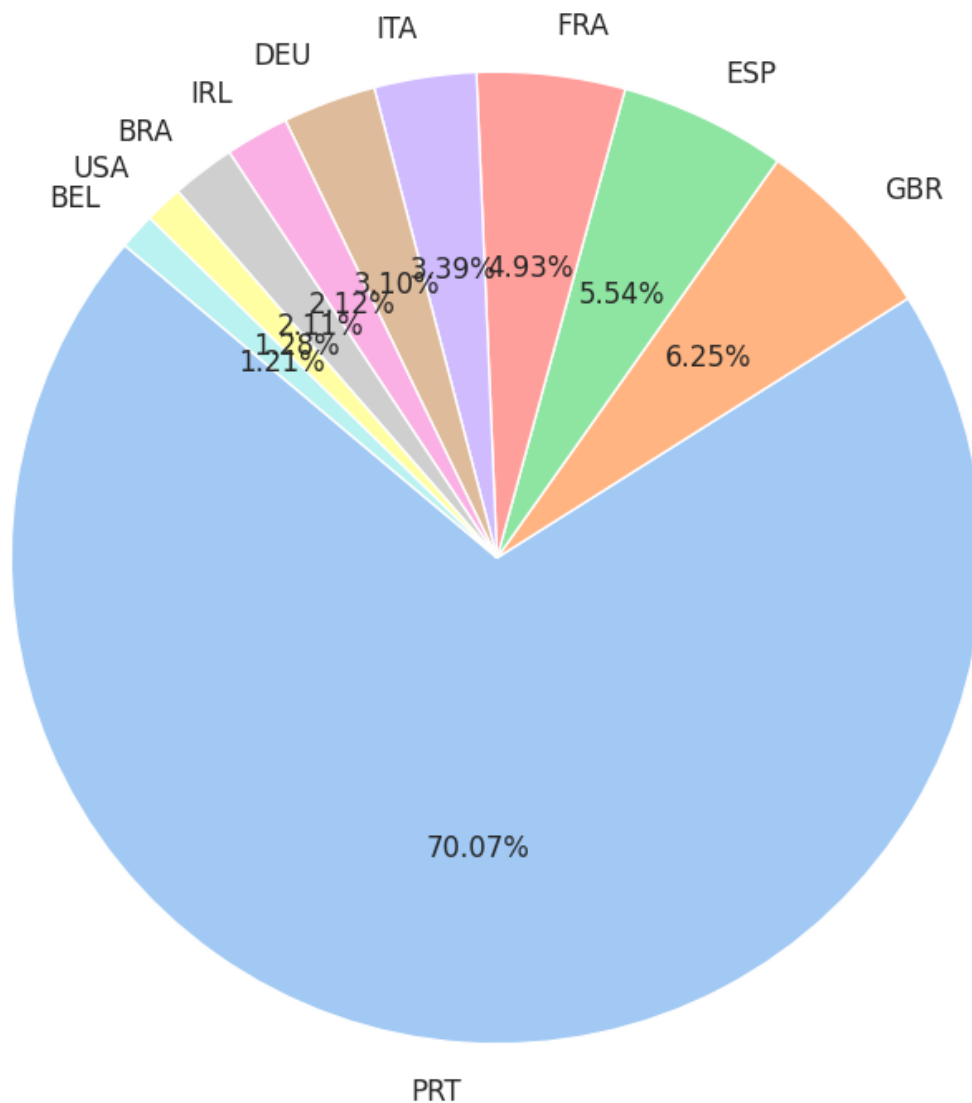
the number of confirmed reservations and the number of canceled reservations are largest in the month of August, whereas January is the month with the most canceled reservations.



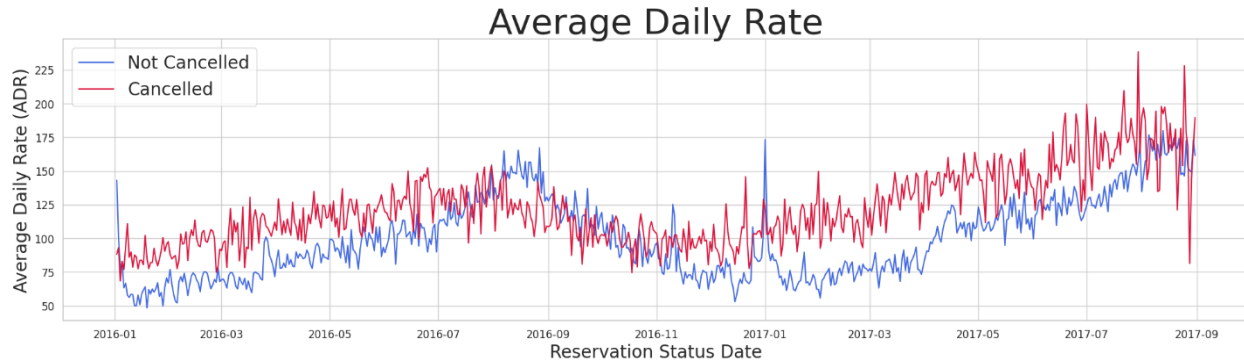
This bar graph demonstrates that cancellations are most common when prices are greatest and are least common when they are lowest. Therefore, the cost of the accommodation is solely responsible for the cancellation.

Now, let's see which country has the highest reservation canceled. The top country is Portugal with the highest number of cancellations.

Top 10 Countries with Reservation Cancellations



Let's check the area from where guests are visiting the hotels and making reservations. Is it coming from Direct or Groups, Online or Offline Travel Agents? Around 46% of the clients come from online travel agencies, whereas 27% come from groups. Only 4% of clients book hotels directly by visiting them and making reservations.



As seen in the graph, reservations are canceled when the average daily rate is higher than when it is not canceled. It clearly proves all the above analysis, that the higher price leads to higher cancellation.

Suggestions

1. Cancellation rates rise as the price does. In order to prevent cancellations of reservations, hotels could work on their pricing strategies and try to lower the rates for specific hotels based on locations. They can also provide some discounts to the consumers.
2. As the ratio of the cancellation and not cancellation of the resort hotel is higher in the resort hotel than the city hotels. So the hotels should provide a reasonable discount on the room prices on weekends or on holidays.
3. In the month of January, hotels can start campaigns or marketing with a reasonable amount to increase their revenue as the cancellation is the highest in this month.
4. They can also increase the quality of their hotels and their services mainly in Portugal to reduce the cancellation rate.

Thank you for reviewing this report.